



Quantitative Finance – ECO 41552-ONL (Spring 2024)
Tue 6 – 7:15pm / Thu 2 – 3:15pm (asynchronous), Jan 25 – May 22

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Final Project Assignment

Draft Due Apr 18, 2024

DUE May 15, 2024

Primary Objective: Use Berry Cox's price momentum factors to select basket of assets for a long-short portfolio. Backtest your algorithm for 5 years to check for performance of algorithm.

Background: The use of computers and coding allows finance professionals to identify new alpha factors – Alpha factors are ways of quantifying where alpha is being generated by an asset. You will be responsible for using Berry Cox's list of alpha factors to determine the long/short baskets for your portfolio. But in order to test your code and its asset selections, back-testing must be performed.

The Requirements:

~~Step 1: Choose an ETF with a minimum of 100 assets, identify those assets~~

~~Step 2: Retrieve historical data for your chosen ETF~~

~~Step 3: Calculate the price momentum factors for each asset in your ETF~~

~~Step 4: Using the price momentum factors, calculate the monthly z-factor score for each asset~~

~~Step 5: Identify long and short baskets (10 to 15 assets in each) using calculated z-factors~~

Step 6: Create a backtest to validate performance of your algorithm based on monthly restructuring over the previous 5 years.

Step 7: Chart:

1. Monthly portfolio return bar chart (pos/neg coloring) vs ETF
2. Monthly return for long picks vs short picks vs ETF
3. Cumulative portfolio return vs ETF

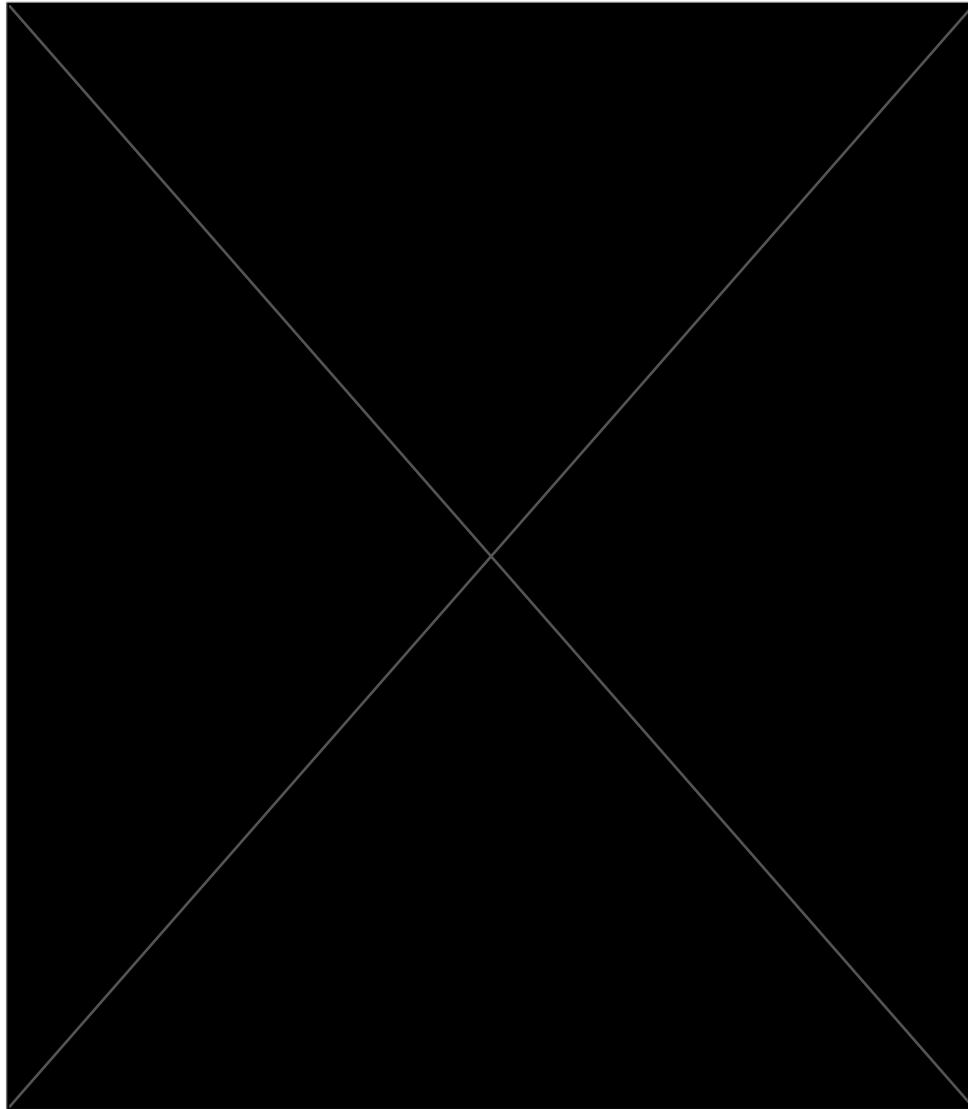
Extra Credit: Calculate the optimized weighting of the monthly restructuring and include in back-testing of your trading algorithm. Make sure the weighting is performed each month when the portfolio is restructured.

Parting Thoughts: Remember, we'll work through all the Python and finance steps you need to understand to be able to calculate alpha factors. That being said, I'm not going to just give you everything – you'll be responsible for determining the actual function for each factor and whether it's a positive or negative factor.

Berry Cox's Alpha Generators are on the next page. I recommend beginning on this project sooner rather than later so you have time to experiment and research how to calculate the specific factors correctly as well as build your backtest.

Good luck and try to have fun!

For final project, find the factors for Price Momentum, since the information used is free. the information to find other factors is not.



Category	Description of Individual Factor	
Price Momentum	Slope of 52 Week Trend Line	(20 Day Lag)
	Percent Above 260 Day Low	(20 Day Lag)
	4/52 Week Price Oscillator	(20 Day Lag)
	39 Week Return	(20 Day Lag)
	51 Week Volume Price Trend	(20 Day Lag)